

Multi-Commodity Military Logistics Optimization

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1 Sets and Network Structure

The optimization model is defined over the following sets.

i Primary set containing all locations in the logistics network. The set consists of Supply Locations (SL), Forward Stations (FS), and Delivery Locations (DL):

$$i = SL \cup FS \cup DL,$$

where

$$|SL| = 3, \quad |FS| = 4, \quad |DL| = 4.$$

j Alias of the location set i , used to denote a second location when defining directed arcs.

SL Subset of i representing the Supply Locations.

FS Subset of i representing the Forward Stations.

DL Subset of i representing the Delivery Locations.

$comt$ Set of commodity types:

$$comt = \{\text{fuel, troops, ammunition, weapons}\}.$$

$arcs$ Set of feasible directed arcs:

$$arcs = \{(i, j) : i \in SL, j \in FS\} \cup \{(i, j) : i \in FS, j \in DL\}.$$

$connectors$ Dynamic set containing the individual transportation connectors. The connector set consists of mode-specific subsets for boats, planes, trains, and trailers:

$$connectors = V^{\text{Boat}} \cup V^{\text{Plane}} \cup V^{\text{Train}} \cup V^{\text{Trailer}}.$$

The cardinality of each subset is determined by the number of vehicles specified for the corresponding experiment.

2 Decision Variables

The optimization model employs the following decision variables.

$\beta_{c,v,i,j}$ Continuous nonnegative variable representing the amount of commodity $c \in \text{comt}$ transported using connector $v \in \text{connectors}$ from location i to location j .

$\alpha_{v,i,j}$ Binary variable indicating whether connector v is assigned to travel from location i to location j .

$\text{total}_{c,i,j}$ Continuous nonnegative variable representing the total amount of commodity c transported from location i to location j across all connectors.

3 Model Parameters

The optimization model uses the following parameters.

cost_v Transportation cost associated with connector v , measured in cost per ton-mile.

maintenance_v Maintenance cost incurred when connector v is used for a trip.

$\text{capacity}_{c,v}$ Maximum amount of commodity c that connector v can transport.

maxCapacity_v Maximum total weight, in tons, that connector v can carry.

wt_c Weight in tons of one unit of commodity c .

The original commodity units are:

- Fuel: U.S. thousand gallons
- Troops: number of personnel
- Ammunition: tons
- Weapons: number of weapons

velocity_v Travel speed of connector v , measured in miles per hour.

$\text{demand}_{c,j}$ Input demand for commodity c at Delivery Location j .

4 Geographical Distance Calculation

Transportation cost and travel time depend on the geographical distance between the locations in the logistics network. Pairwise distances are computed from the latitude and longitude coordinates of the 11 locations using the Haversine calculation implemented in the model.

Let location i have latitude ϕ_i and longitude λ_i , and location j have latitude ϕ_j and longitude λ_j . Define

$$\Delta\phi_{ij} = \phi_j - \phi_i, \tag{1}$$

$$\Delta\lambda_{ij} = \lambda_j - \lambda_i. \tag{2}$$

The Haversine term is

$$a_{ij} = \sin^2 \left(\frac{\Delta\phi_{ij}}{2} \right) + \cos(\phi_i) \cos(\phi_j) \sin^2 \left(\frac{\Delta\lambda_{ij}}{2} \right), \quad (3)$$

with

$$c_{ij} = 2 \tan^{-1} \left(\frac{\sqrt{a_{ij}}}{\sqrt{1 - a_{ij}}} \right). \quad (4)$$

The distance parameter used by the implementation is

$$d_{ij} = \frac{R c_{ij}}{1.6}, \quad (5)$$

where the implementation specifies

$$R = 3982.$$

The resulting values are stored in the parameter

$$d_{ij}, \quad i, j \in i.$$

This distance parameter is subsequently used in the transportation-cost and travel-time calculations.

5 Deterministic Cost-Minimization Model

The deterministic model determines a minimum-cost transportation plan subject to the operational restrictions implemented in the GAMSPy model.

5.1 Connector Restrictions

Plane Fuel Restriction Aircraft are prohibited from transporting fuel:

$$\sum_{v \in V^{\text{Plane}}} \beta_{\text{fuel}, v, i, j} = 0, \quad \forall (i, j) \in \text{arcs}. \quad (6)$$

Train Routing Restriction Trains are prohibited on arcs connecting Forward Stations to Delivery Locations:

$$\sum_{v \in V^{\text{Train}}} \beta_{c, v, i, j} = 0, \quad \forall c \in \text{comt}, i \in FS, j \in DL. \quad (7)$$

Boat Routing Restriction Boats are prohibited on arcs connecting Supply Locations to Forward Stations:

$$\sum_{v \in V^{\text{Boat}}} \beta_{c, v, i, j} = 0, \quad \forall c \in \text{comt}, i \in SL, j \in FS. \quad (8)$$

5.2 Connector Capacity Constraint

The total commodity weight assigned to a connector on an arc cannot exceed the maximum carrying capacity of the connector:

$$\sum_{c \in comt} wt_c \beta_{c,v,i,j} \leq maxCapacity_v, \quad \forall v \in connectors, (i,j) \in arcs. \quad (9)$$

5.3 Water Accessibility Constraints

The parameter ω represents the value of **waterBwDL**:

$$\omega = \begin{cases} 1, & \text{if a water route exists between } FS \text{ and } DL, \\ 0, & \text{otherwise.} \end{cases}$$

When $\omega = 0$, boat commodity flow on $FS \rightarrow DL$ arcs is fixed to zero:

$$\beta_{c,v,i,j} = 0, \quad \forall c \in comt, v \in V^{Boat}, i \in FS, j \in DL. \quad (10)$$

When $\omega = 1$, trains and trailers are prohibited on $FS \rightarrow DL$ arcs:

$$\alpha_{v,i,j} = 0, \quad \forall v \in V^{Train}, i \in FS, j \in DL, \quad (11)$$

$$\alpha_{v,i,j} = 0, \quad \forall v \in V^{Trailer}, i \in FS, j \in DL. \quad (12)$$

Independently of ω , boats are prohibited on all $SL \rightarrow FS$ arcs:

$$\alpha_{v,i,j} = 0, \quad \forall v \in V^{Boat}, i \in SL, j \in FS. \quad (13)$$

5.4 Commodity Source Constraints

The implementation restricts each commodity to its designated Supply Location.

Commodity	Supply Location
Fuel	Houston, Texas (SL3)
Troops	New Orleans, Louisiana (SL2)
Ammunition	Hawthorne, Nevada (SL1)
Weapons	Hawthorne, Nevada (SL1)

The corresponding restrictions are

$$total_{fuel,SL1,j} = 0, \quad \forall j \in FS, \quad (14)$$

$$total_{fuel,SL2,j} = 0, \quad \forall j \in FS, \quad (15)$$

$$total_{troops,SL1,j} = 0, \quad \forall j \in FS, \quad (16)$$

$$total_{troops,SL3,j} = 0, \quad \forall j \in FS, \quad (17)$$

$$total_{ammunition,SL2,j} = 0, \quad \forall j \in FS, \quad (18)$$

$$total_{ammunition,SL3,j} = 0, \quad \forall j \in FS, \quad (19)$$

$$total_{weapons,SL2,j} = 0, \quad \forall j \in FS, \quad (20)$$

$$total_{weapons,SL3,j} = 0, \quad \forall j \in FS. \quad (21)$$

5.5 Forward-Station Flow Condition

The implemented Forward-Station condition requires aggregate inflow from the Supply Locations minus aggregate outflow to the Delivery Locations to be nonpositive:

$$\sum_{c \in comt} \sum_{v \in connectors} \sum_{i \in SL} \beta_{c,v,i,f} - \sum_{c \in comt} \sum_{v \in connectors} \sum_{j \in DL} \beta_{c,v,f,j} \leq 0, \quad \forall f \in FS. \quad (22)$$

Thus, as implemented,

$$\text{aggregate inflow at } f \leq \text{aggregate outflow from } f.$$

5.6 Demand Satisfaction Constraints

For each commodity, total flow from Supply Locations to Forward Stations must cover the aggregate demand across the Delivery Locations:

$$\sum_{i \in SL} \sum_{f \in FS} total_{c,i,f} \geq \sum_{j \in DL} demand_{c,j}, \quad \forall c \in comt. \quad (23)$$

At each Delivery Location, incoming flow must satisfy the corresponding commodity demand:

$$\sum_{f \in FS} total_{c,f,j} \geq demand_{c,j}, \quad \forall c \in comt, j \in DL. \quad (24)$$

5.7 Flow Aggregation Constraint

The aggregate flow variable is defined as

$$total_{c,i,j} = \sum_{v \in connectors} \beta_{c,v,i,j}, \quad \forall c \in comt, (i,j) \in arcs. \quad (25)$$

5.8 Connector Activation Constraint

Commodity flow on a connector is bounded by its commodity-specific capacity and can be positive only when the connector is activated:

$$\beta_{c,v,i,j} \leq capacity_{c,v} \alpha_{v,i,j}, \quad \forall c \in comt, v \in connectors, (i,j) \in arcs. \quad (26)$$

where

$$\alpha_{v,i,j} \in \{0, 1\}.$$

5.9 Connector Availability Constraint

For every connector and origin location, the connector can be assigned to at most one destination:

$$\sum_{j \in i} \alpha_{v,i,j} \leq 1, \quad \forall v \in \text{connectors}, i \in i. \quad (27)$$

5.10 Objective Function

The total cost is defined as

$$\begin{aligned} TotalCost = & \sum_{c \in comt} \sum_{v \in connectors} \sum_{(i,j) \in arcs} \beta_{c,v,i,j} wt_c cost_v d_{ij} \\ & + \sum_{v \in connectors} \sum_{i \in i} \sum_{j \in i} maintenance_v \alpha_{v,i,j}. \end{aligned} \quad (28)$$

The deterministic model minimizes

$$\min TotalCost. \quad (29)$$

6 Minimum-Time Model

The minimum-time model retains the previously defined operational constraints and introduces the time variables used by the GAMSPy implementation.

6.1 Decision Variables

$t_1(i, f)$ First-stage time variable for Supply Location i and Forward Station f .

$t_2(f, j)$ Second-stage time variable for Forward Station f and Delivery Location j .

T Scalar variable `time` representing the model's total-time value.

6.2 Travel-Time Constraints

For every connector on the first stage,

$$t_1(i, f) \geq \frac{d_{if}}{velocity_v} \alpha_{v,i,f}, \quad \forall v \in \text{connectors}, i \in SL, f \in FS. \quad (30)$$

For every connector on the second stage,

$$t_2(f, j) \geq \frac{d_{fj}}{velocity_v} \alpha_{v,f,j}, \quad \forall v \in \text{connectors}, f \in FS, j \in DL. \quad (31)$$

The GAMSPy implementation defines the scalar time variable using an equality for every (i, f, j) combination:

$$T = t_1(i, f) + t_2(f, j), \quad \forall i \in SL, f \in FS, j \in DL. \quad (32)$$

6.3 Objective Function

The objective is

$$\min T. \tag{33}$$

7 Time-Constrained Commodity Flow Maximization

This formulation uses the same two-stage network while treating the delivered quantity as a decision variable. Ten connectors of each transportation mode are used in this model.

7.1 Connector Availability

$$|ct1| = |ct2| = |ct3| = |ct4| = 10,$$

so that

$$|connectors| = 40.$$

7.2 Modified Demand Variable

The variables

$$\alpha_{v,i,j}, \quad \beta_{c,v,i,j}, \quad total_{c,i,j}$$

retain their previous definitions.

In this formulation,

$$demand_{c,j} \geq 0$$

is declared as a decision variable.

7.3 Inherited Operational Constraints

The connector restrictions, connector-capacity constraints, water accessibility restrictions, commodity-source restrictions, aggregate Forward-Station condition, flow aggregation, connector activation, and connector availability conditions are retained from the deterministic model.

Because **demand** is now a variable, the demand relations are

$$\sum_{i \in SL} \sum_{f \in FS} total_{c,i,f} \geq \sum_{j \in DL} demand_{c,j}, \quad \forall c \in comt, \tag{34}$$

and

$$\sum_{f \in FS} total_{c,f,j} \geq demand_{c,j}, \quad \forall c \in comt, j \in DL. \tag{35}$$

7.4 Time Definition and Time Limit

The time variables satisfy the same arc-level inequalities as in the minimum-time model:

$$t_1(i, f) \geq \frac{d_{if}}{velocity_v} \alpha_{v,i,f}, \quad (36)$$

and

$$t_2(f, j) \geq \frac{d_{fj}}{velocity_v} \alpha_{v,f,j}. \quad (37)$$

The implementation again defines

$$T = t_1(i, f) + t_2(f, j), \quad \forall i \in SL, f \in FS, j \in DL. \quad (38)$$

For the time-constrained flow experiment, the completion time is bounded by the prescribed limit:

$$T \leq \bar{T}. \quad (39)$$

7.5 Commodity-Flow Maximization Objective

For the implemented commodity-flow experiment, the quantity of the selected commodity delivered to the Delivery Locations is maximized. For the ammunition experiment,

$$\max \sum_{j \in DL} demand_{\text{ammunition},j}. \quad (40)$$

The corresponding time limit used for this experiment is

$$T \leq 15. \quad (41)$$

8 Minimum-Time Delivery of Fuel

The next formulation uses the same network and operational constraints but considers only the prescribed fuel-demand requirement at the Delivery Locations.

The time variables retain their implemented definitions:

$$T = t_1(i, f) + t_2(f, j), \quad \forall i \in SL, f \in FS, j \in DL. \quad (42)$$

The objective is

$$\min T. \quad (43)$$

Thus, this model returns the minimum value of the implemented total-time variable for the specified fuel-demand instance.

9 Pareto Analysis Under Connector Availability Constraints

The Pareto analysis examines the relationship between transportation cost and the implemented total-time measure under different connector availability levels.

9.1 Connector Availability

For the illustrated experiment, 17 connectors of each mode are used:

$$ct1 = \{\text{boat}_1, \dots, \text{boat}_{17}\}, \quad (44)$$

$$ct2 = \{\text{plane}_1, \dots, \text{plane}_{17}\}, \quad (45)$$

$$ct3 = \{\text{train}_1, \dots, \text{train}_{17}\}, \quad (46)$$

$$ct4 = \{\text{trailer}_1, \dots, \text{trailer}_{17}\}. \quad (47)$$

Therefore,

$$|ct1| = |ct2| = |ct3| = |ct4| = 17,$$

and

$$|\text{connectors}| = 68.$$

9.2 Modified Demand Data

The Pareto experiment uses demand equal to twice the deterministic demand.

Let $\overline{D}_{c,j}$ denote the deterministic demand. Then

$$D_{c,j}^P = 2\overline{D}_{c,j}, \quad \forall c \in \text{comt}, j \in DL. \quad (48)$$

9.3 Inherited Constraints

The connector restrictions, capacity constraints, Haversine distance calculation, water-accessibility conditions, commodity-source restrictions, Forward-Station condition, demand constraints, flow aggregation, connector-activation constraint, and connector-availability constraint retain the same form as in the deterministic model.

The time variables are defined as

$$t_1(i, f) \geq \frac{d_{if}}{\text{velocity}_v} \alpha_{v,i,f}, \quad (49)$$

$$t_2(f, j) \geq \frac{d_{fj}}{\text{velocity}_v} \alpha_{v,f,j}, \quad (50)$$

and

$$T = t_1(i, f) + t_2(f, j), \quad \forall i \in SL, f \in FS, j \in DL. \quad (51)$$

9.4 Cost Minimization Subject to a Time Bound

A parameter $T^{\max}$ is used to impose

$$T \leq T^{\max}. \quad (52)$$

The model minimizes

$$\min \text{TotalCost}. \quad (53)$$

The parameter is initially assigned a value of 100 in the model construction, and is subsequently overwritten during Pareto generation.

For the Pareto experiment,

$$T^{\max} \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}.$$

9.5 Pareto Frontier Computation

For every value of $T^{\max}$, the model is solved as a mixed-integer cost-minimization problem. If a feasible integer solution is returned, the resulting total cost and implemented time value are recorded.

Let

$$J_1 = TotalCost$$

and

$$J_2 = T.$$

The recorded pairs

$$(J_1, J_2)$$

form the cost–time curve for the specified connector availability level.

The same procedure is repeated for different numbers of available connectors to compare the corresponding cost–time relationships.

10 Expected-Cost Optimization Under Stochastic Demand

Demand uncertainty is represented using five explicit scenarios.

10.1 Scenario Structure

For this model, 25 connectors of each transportation mode are available:

$$|ct1| = 25, \tag{54}$$

$$|ct2| = 25, \tag{55}$$

$$|ct3| = 25, \tag{56}$$

$$|ct4| = 25, \tag{57}$$

and therefore

$$|connectors| = 100. \tag{58}$$

The scenario set is

$$S = \{0, 1, 2, 3, 4\}. \tag{59}$$

10.2 Scenario-Indexed Decision Variables

The decision variables are extended by a scenario index:

$$\alpha_{v,i,j,s} \in \{0, 1\},$$

$$\beta_{c,v,i,j,s} \geq 0,$$

and

$$total_{c,i,j,s} \geq 0.$$

Demand is scenario-dependent and is represented as

$$D_{c,j}^s.$$

The transportation parameters

$$cost_v, maintenance_v, capacity_{c,v}, maxCapacity_v, wt_c, velocity_v, d_{ij}$$

are not scenario indexed.

10.3 Scenario Probabilities

The scenario probabilities are

$$(p_0, p_1, p_2, p_3, p_4) = (0, 0.25, 0.25, 0.25, 0.25),$$

with

$$\sum_{s \in S} p_s = 1. \quad (60)$$

10.4 Scenario Operational Constraints

The deterministic connector, capacity, water-accessibility, commodity-source, flow, demand, aggregation, activation, and connector-use constraints are reproduced for each scenario s .

For example, the implemented Forward-Station condition becomes

$$\sum_{c \in comt} \sum_{v \in connectors} \sum_{i \in SL} \beta_{c,v,i,f,s} - \sum_{c \in comt} \sum_{v \in connectors} \sum_{j \in DL} \beta_{c,v,f,j,s} \leq 0, \quad \forall f \in FS, s \in S. \quad (61)$$

The demand constraints become

$$\sum_{i \in SL} \sum_{f \in FS} total_{c,i,f,s} \geq \sum_{j \in DL} D_{c,j}^s, \quad \forall c \in comt, s \in S, \quad (62)$$

and

$$\sum_{f \in FS} total_{c,f,j,s} \geq D_{c,j}^s, \quad \forall c \in comt, j \in DL, s \in S. \quad (63)$$

10.5 Scenario Travel-Time Variables

The time variables are also scenario indexed:

$$t_1(i, f, s), \quad t_2(f, j, s), \quad T_s.$$

They satisfy

$$t_1(i, f, s) \geq \frac{d_{if}}{\text{velocity}_v} \alpha_{v,i,f,s}, \quad (64)$$

$$t_2(f, j, s) \geq \frac{d_{fj}}{\text{velocity}_v} \alpha_{v,f,j,s}, \quad (65)$$

and the implemented time definition is

$$T_s = t_1(i, f, s) + t_2(f, j, s), \quad \forall i \in SL, f \in FS, j \in DL, s \in S. \quad (66)$$

10.6 Expected-Cost Objective

The objective implemented in GAMSPy minimizes the probability-weighted transportation and connector-maintenance cost:

$$\min \sum_{s \in S} p_s \left[\sum_{c \in \text{comt}} \sum_{v \in \text{connectors}} \sum_{(i,j) \in \text{arcs}} \beta_{c,v,i,j,s} \text{wt}_c \text{cost}_v d_{ij} + \sum_{v \in \text{connectors}} \sum_{i \in i} \sum_{j \in i} \text{maintenance}_v \alpha_{v,i,j,s} \right]. \quad (67)$$

The implementation solves this model as a mixed-integer program using XPRESS with relative optimality gap

0.12.

Because α , β , and $total$ are all scenario indexed, each scenario has its own transportation decisions in the implemented formulation.

11 Minimax Optimization Under Stochastic Demand

The minimax model uses the same five scenario-specific formulations but replaces the expected-cost objective with the maximum scenario cost.

11.1 Scenario Cost

For each scenario s , define

$$\begin{aligned} C_s = & \sum_{c \in \text{comt}} \sum_{v \in \text{connectors}} \sum_{(i,j) \in \text{arcs}} \beta_{c,v,i,j,s} \text{wt}_c \text{cost}_v d_{ij} \\ & + \sum_{v \in \text{connectors}} \sum_{i \in i} \sum_{j \in i} \text{maintenance}_v \alpha_{v,i,j,s}. \end{aligned} \quad (68)$$

11.2 Maximum-Cost Variable

The variable $MaxCost$ satisfies

$$MaxCost \geq C_s, \quad \forall s \in S. \quad (69)$$

11.3 Minimax Objective

The objective is

$$\min MaxCost. \quad (70)$$

Thus, the implemented minimax formulation minimizes the largest cost among the five scenario-specific transportation solutions.